Association between COVID-19 and residential mobility index in São Paulo, Brazil

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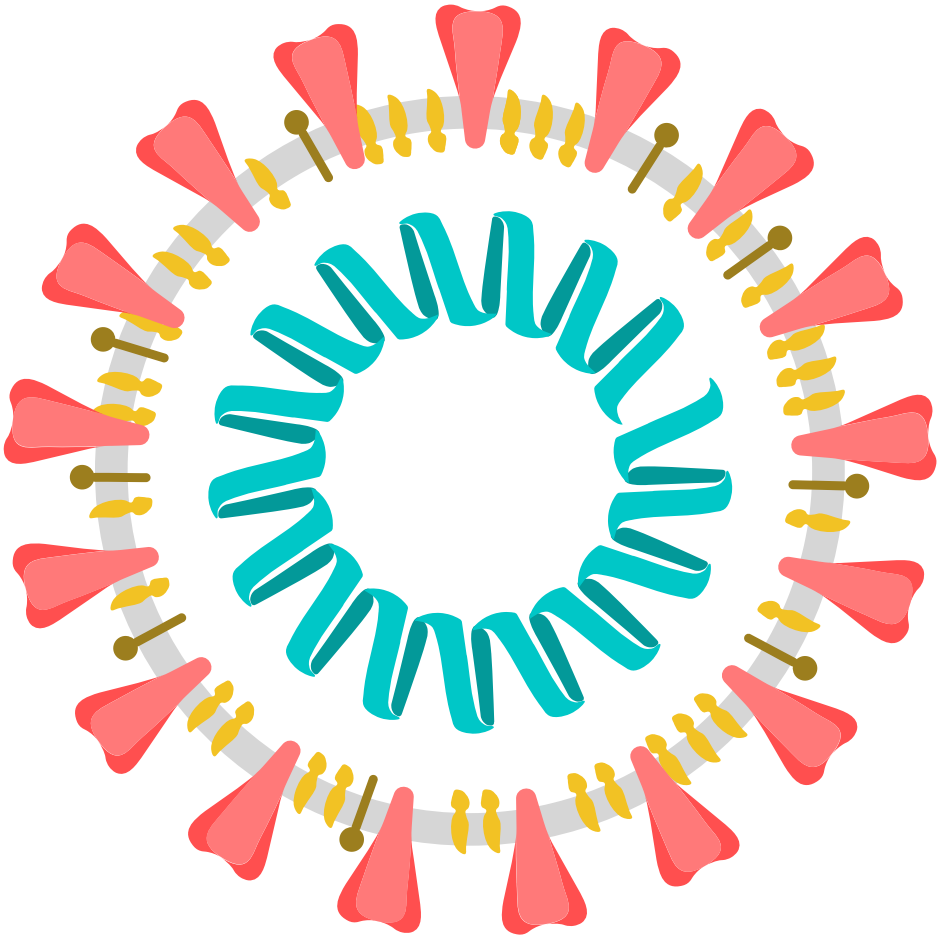
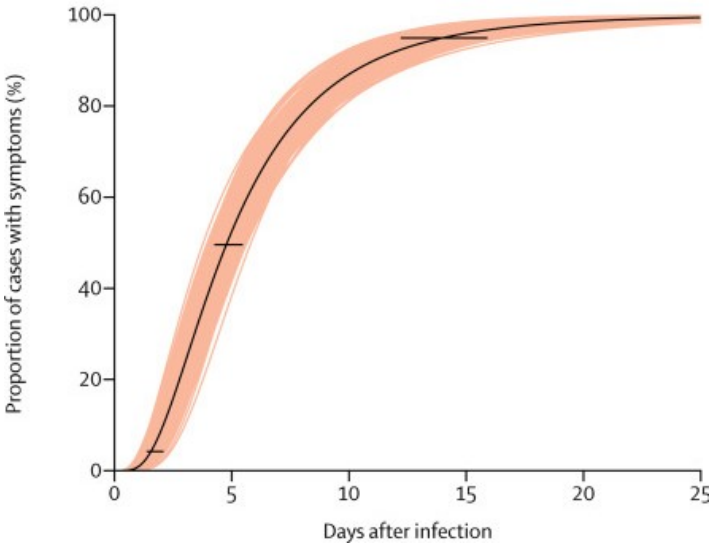


Introduction

1. Introduction - COVID-19[1].

- On December 31, it was reported a cluster of cases of pneumonia in Wuhan, Hubei Province, China.
- 71 days after, the Coronavirus disease **COVID-19**, caused by the **SARS CoV2** virus, was declared as Pandemic on March 11, 2020.

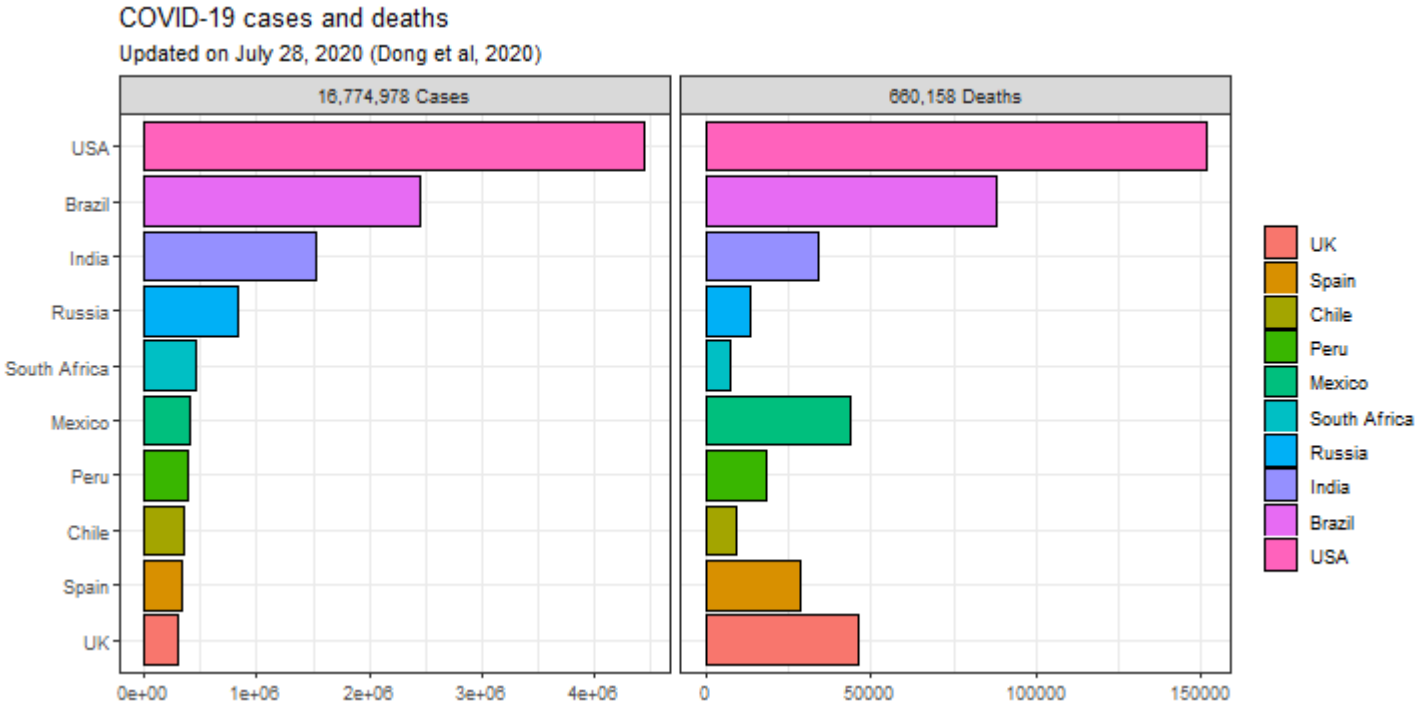


Incubation time

- Time from infection to develop symptoms
- Median of 5 days.
- 95% of symptomatic cases develop symptoms will within 0-14 days of infection.
- (Qifang et al, 2020)

- Internet companies such as Google and Apple released GPS data.

Top 10 countries with cases and deaths of COVID-19



- It is imperative to study environment and COVID-19 (Qu et al., 2020).

Brazil

1. Brazil has a population of 208 million inhabitants.
2. The state of São Paulo has 44 million inhabitants; 20% of cases and 25% of deaths.
3. Only symptomatic cases are tested. Each state defines quarantine measures.
4. During the São Paulo quarantine, CO ↓ -35.8%, NO_2 -26.6%, PM_{10} ↓ 17.8%, $PM_{2.5}$ ↓ 25.0% and O_3 ↑ 53% (Freitas et al., 2020).

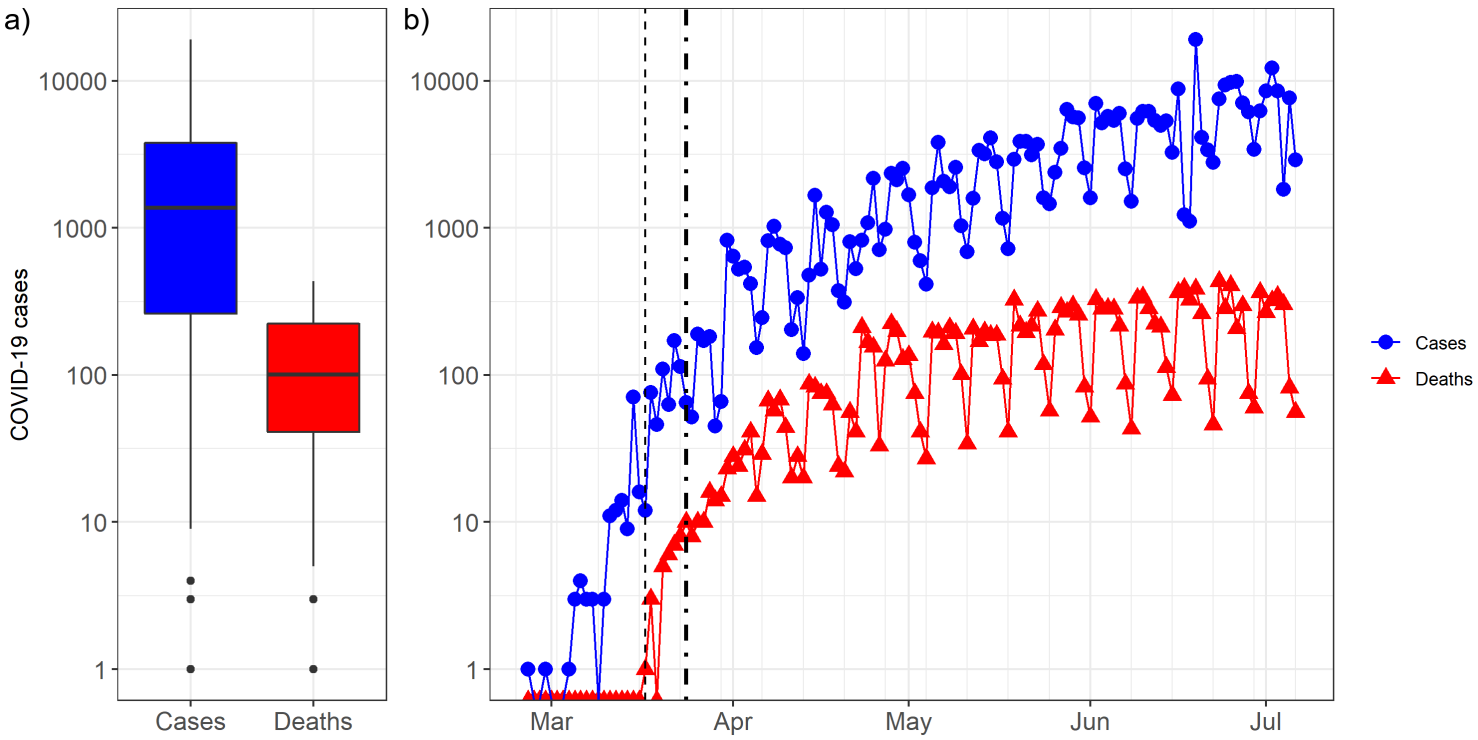
- source

Objetives

To analyze the effects of quarantine, air pollution, and meteorology on the daily cases and deaths of COVID-19 in the state of Sao Paulo.

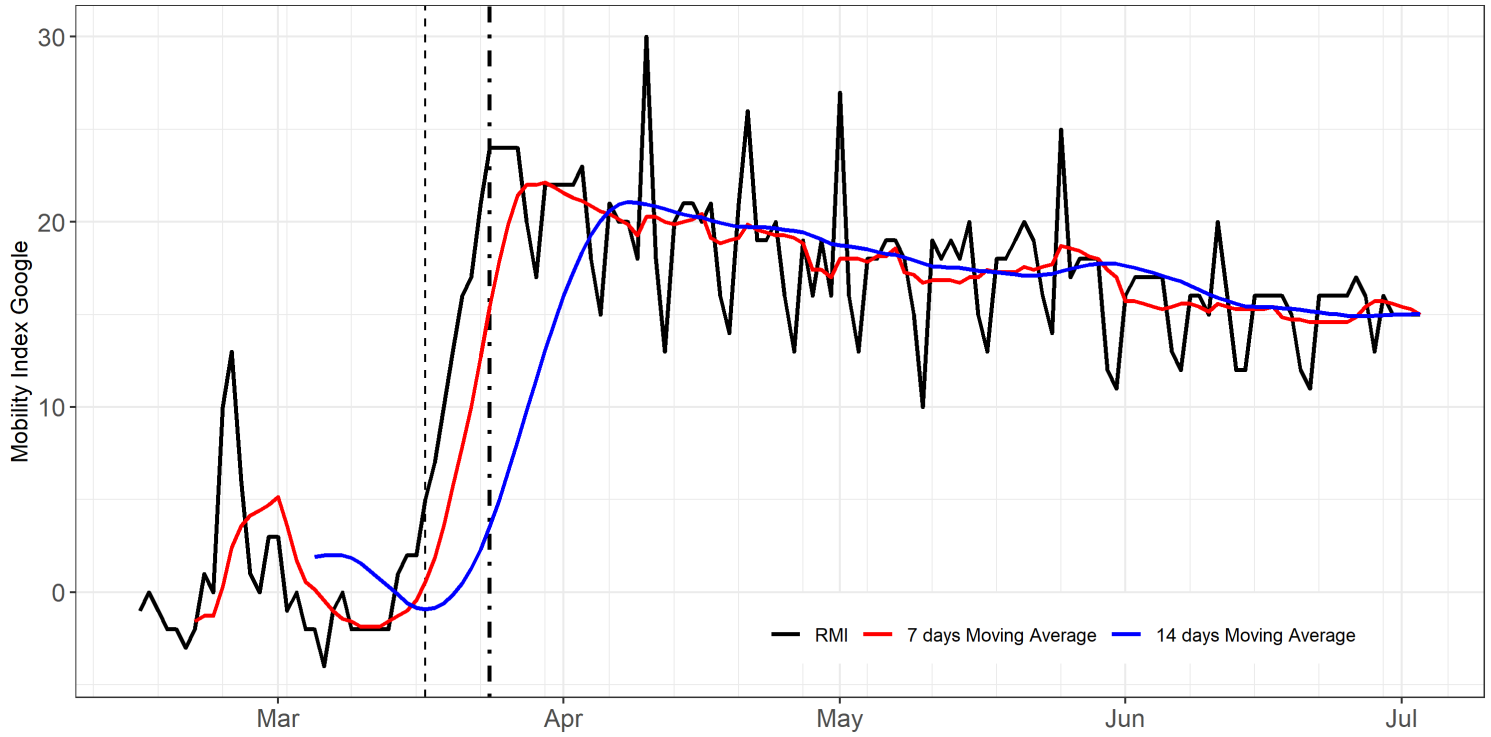
Data

Cases and mortality in São Paulo state



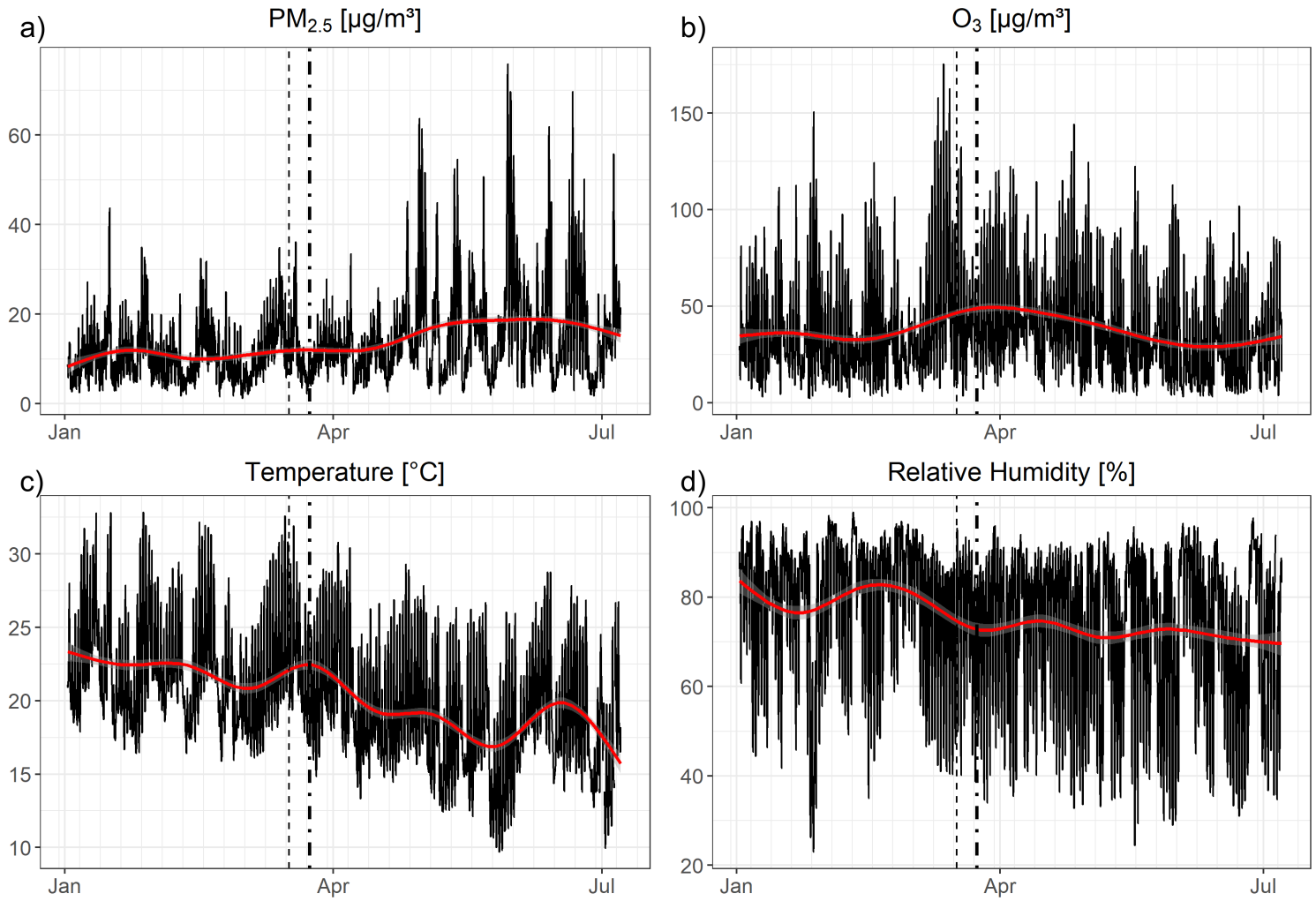
- The first case on February 26th, first death on March 17th, 2020.
- The mean of daily cases is 2342, maximum of 19030 on June 19th, 2020.
- The mean daily deaths is 119, maximum of 434 on June 23th, 2020.

Residential mobility index (RMI, Google)



- The spatial distancing was decreed on March 17, quarantine on March 24.
- There is a trend to return to previous quarantine period.
- It is expected that RMI would reach a value of 10% on August 8th, 2020
- Average RMI during June was 15.30%

Environmental parameters



Methodology

Effects of RMI on COVID-19

- Semiparametric generalized additive model. This approach has been used in health effect of air pollution studies (Dominici et al., 2004; Peng & Dominici, 2008).
- 7-14 days moving average of variables.

$$log(u) = \beta_0 + \beta_1 * RMI_m + s(temp_m) + s\left(PM_{2.5_m}\right) + s(time) + s(dow) + s(\log(u_{-1}))$$

$$log(u) = \beta_0 + \beta_1 * RMI_m + s(temp_m) + s\left(O_{3_m}\right) + s(time) + s(dow) + s(\log(u_{-1}))$$

$$log(u) = \beta_0 + \beta_1 * RMI_m + s(rh_m) + s\left(PM_{2.5_m}\right) + s(time) + s(dow) + s(\log(u_{-1}))$$

$$log(u) = \beta_0 + \beta_1 * RMI_m + s(rh_m) + s\left(O_{3_m}\right) + s(time) + s(dow) + s(\log(u_{-1}))$$

- Final model based on Akaike Information Criterion (Sakamoto et al., 1986).

Effects of Air pollution on COVID-19

$$log(u) = \beta_0 + \beta_1 * P_m + s(RMI_m) + s(temp_m) + s(time) + s(dow) + s(\log(u_{-1}))$$

$$log(u) = \beta_0 + \beta_1 * P_m + s(RMI_m) + s(RH_m) + s(time) + s(dow) + s(\log(u_{-1}))$$

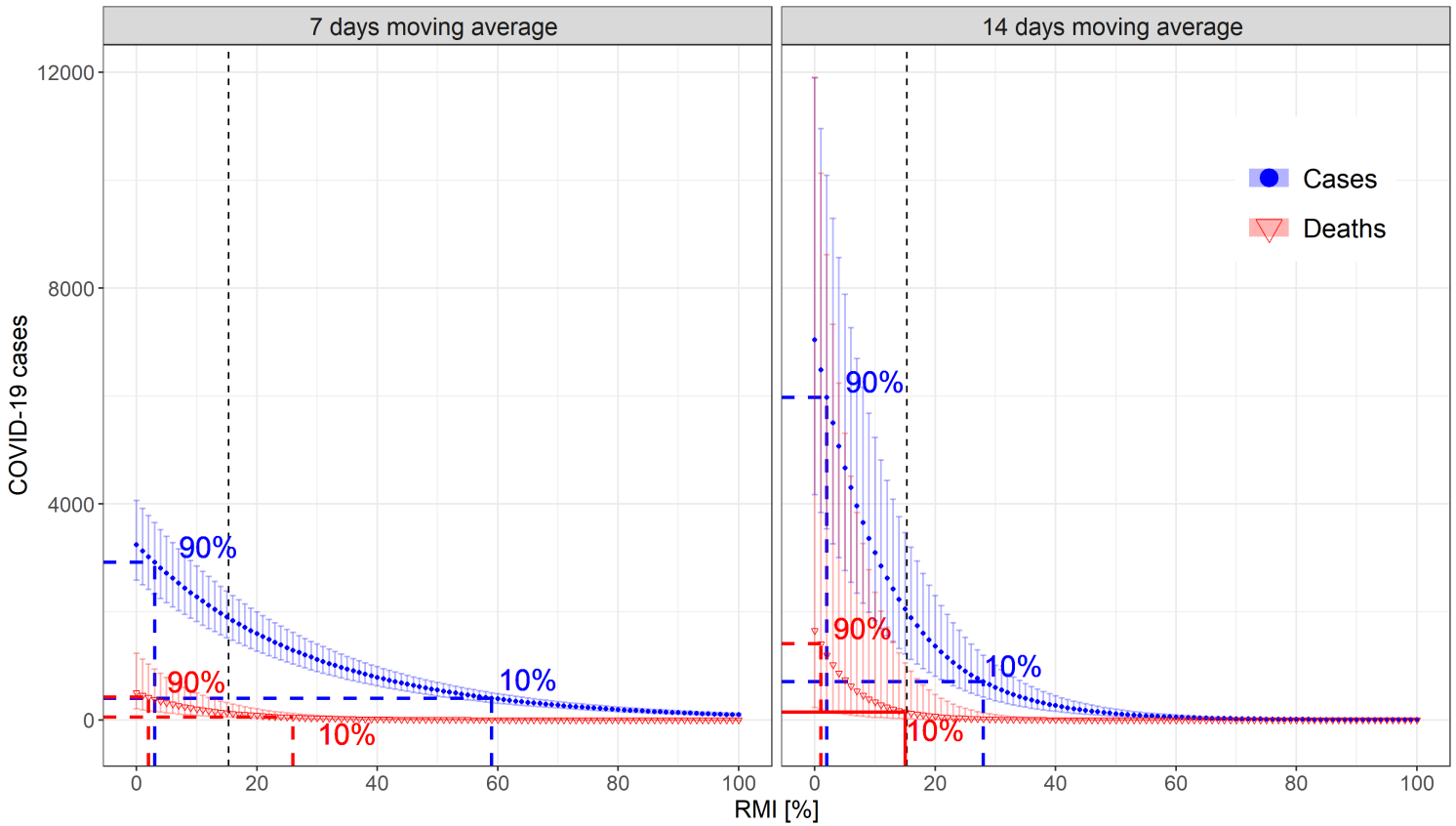
- Where, P is air pollutant concentration moving average of $PM_{2.5}$ or O_3.
- met are the moving averge of daily averages of temperature and relative humidity.

Then, the relative risk is calculated as

$$exp(\beta * X)$$

Results

Effect of social distancing on COVID-19



- 1887 cases (CI: 1507 to 2362)
- 125 deaths (CI: 51 to 309)

- 2004 cases (CI: 1187 to 3384)
- 140 deaths (CI: 19 to 1005)

Extreme and critical RMI values

7 days moving average

RMI 0%

- 3244 cases (CI: 2591 to 4061)
- 501 deaths (CI: 203 to 1234)

RMI 100%

- 94 cases (CI: 75 to 118)
- 0.059 deaths (CI: 0.024 to 0.145)

Critical RMI

- 3% for cases and 2%

14 days moving average

RMI 0%

- 7045 (CI: 4173 to 11893)
- 1656 deaths (CI: 230 to 11904)

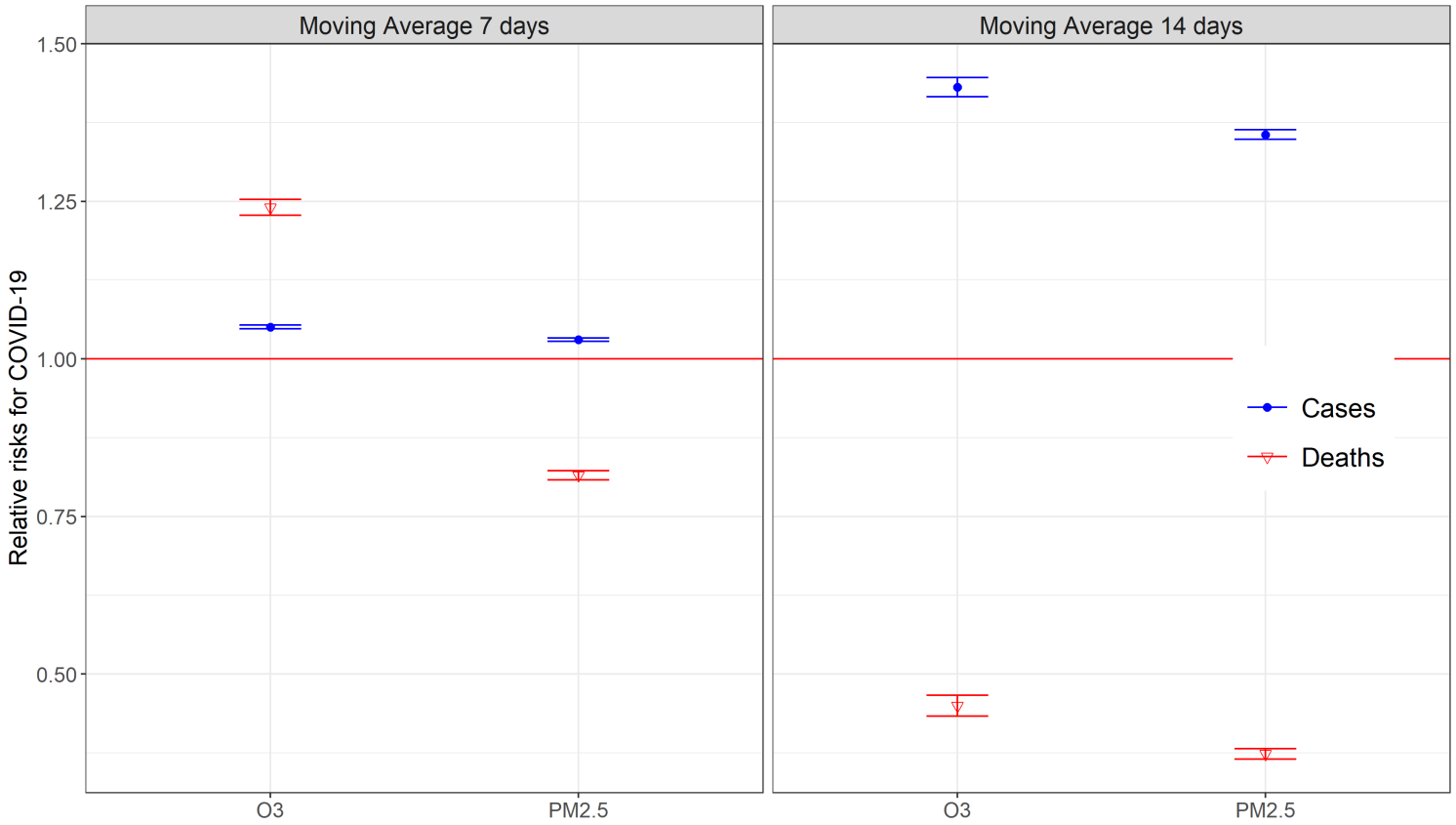
RMI 100%

- 2 cases (CI: 1 to 3)
- 1.6e-04 deaths (CI: 2e-05 to 1.15e-03)

Critical RMI

- 2%

Effect of increment air pollutions on COVID-19



Discussion

Discussion

- We found statistical associations between RMI and COVID-19 cases
- A higher RMI (more stay-at-home) results in lower COVID-19 cases, and less RMI (more outdoor activities) resulted in many more cases of COVID-19.
- The median and mean for cases and deaths related to COVID-19 in São Paulo are 1111 and 2429, and 75 and 121, respectively, which means that our predictions have a good agreement with observations
- Zhu et al., (2020) found that 10 $\mu g/m^3$ results in relative risk of 1.02 after 14 days, which is similar with the value of and we 1.03 presented in this study after 7 days.
- Mobility indices can be used as a tool by policymakers to implement more effective social distancing and quarantine-based management strategies for COVID-19.

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Obrigado!